

INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 6: Conclusion and Future Work

by

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6 Conclusion and Future Work

In Chapter 5 the findings were interpreted and whether the study’s objectives had been met assessed. The tables in Chapter 4 are the record.

6.1 Answers to the research questions

This thesis proposed EndorseRank, a version of PageRank operating on a latest-allowance endorsement graph, and compared it to Adaptive Weighted PageRank (AWP), the transfer-graph baseline defined by Do, Do and Nguyen^[3], using a single cohort of Arbitrum One wallets matched to their spenders, a single cohort of holdouts representing potential future approvers and a single pool of expanded addresses spanning December 2025 through May 2026.

RQ1 investigated whether EndorseRank and AWP provide significantly different rankings for wallets within the same sample. They do, as measured by the inter-method rank agreement (Table 4.3), which falls between the values expected for independent and perfectly correlated orderings, excluding both those limits, and the scatter plot of ranks shows significant disagreement between the two methods. Thus hypothesis H1 is validated: authorization graphs and payment graphs represent separate reputation landscapes.

RQ2 investigated which method exhibits greater alignment with allowance transactions and with transfer transactions within the same timeframe. The response depends on the construct. AWP is closer to transfer proxies and EndorseRank is closer to allowance proxies, and the two between-method paired differences are non-zero and in opposite directions (Table 4.4), so H2 is true in this form. The within-method differences (EndorseRank’s allowance closeness versus its own transfer closeness) are not significant, so the thesis does not conclude that EndorseRank prefers one signal to another within its own ranking.

RQ3 asks whether EndorseRank is more computationally efficient than AWP on the same matched sample. It is in terms of wall-clock time, memory, and iterations, owing to the endorsement graph having approximately one-ninth the number of edges of the transfer graph under a single timing protocol (Table 4.1), so H3 is true. The sensitivity to node-counts on the expanded pool verifies the ratio of edge-counts over staged node-counts, but the lack of completion of the second-tier extraction means it cannot test edge-growth (Table 4.2).

RQ4 asks whether scores at t_1 that have been frozen to the time of calculation predict owner-spender approvals at t_2 on inbound spenders more strongly than AWP and more informatively than the same-window allowance degree. The former holds and the latter does not (Tables 4.13 and 4.14). EndorseRank’s interval excludes zero and its paired difference to AWP is far above zero, and the labels post-date every scored event; however, the spender’s raw in-approve degree at t_1 predicts the same labels better than EndorseRank, and the transfer in-degree predicts new senders better than AWP. Propagation separates the two constructs out-of-window, but does not add any predictive value over the local count for these labels. RQ4 replaces the comparison of trading success made in the research proposal which was discarded as circular (Section 2.8); the holdouts test the temporal predictive validity of the endorsement construct relative to the transfer construct, not their credit-risk validity.

RQ5 asks whether a coupled PageRank over both edge types ranks wallets better than either single-layer method. It does not. Under a rule established before the run, C-PR at each tested λ lies between EndorseRank and AWP on both holdout labels and the same-window families, and the seeded ablation reproduces AWP (Section 4.7; Table 4.14). The allowance edges contribute via propagation along them, not via seeding, and their mixing with transfer edges at the transition level dilutes them.

Claims C1 through C5 (Section 4.9) describe an efficiency and alignment trade-off with two bounds. EndorseRank has lower PageRank cost, alignment with the allowance construct, and separation from AWP on future approvals. AWP has stronger alignment with transfer and longitudinal stability. No PageRank outperforms its own raw degree out-of-window, and coupling the two edge types in

one walk outperforms neither single-layer method. The choice of deployment depends on whether the user's query relates to authorization or to flow, and the two scores should remain separate.

6.2 Original contribution to knowledge

Section 1.9 lists four contributions. In the language of the literature reviewed in Chapter 2, this thesis establishes the following.

Prior to this work, graph-reputation research on public ledgers has either ranked wallets on transfers ^{[3],[4]}, ranked them according to how risk propagates through transaction networks ^[12], or assigned them explicit ratings and stakes ^{[25],[26]}. And, in all cases, researchers have checked whether their rankings were valid by comparing them to statistics derived from the input data. No prior work on a public ledger had used the ERC-20 allowance, the on-chain action that encodes a conscious decision to delegate spending power ^[18], as an edge in a graph-reputation calculation, nor had anyone made a same-cohort comparison, complete with confidence intervals, difference tests and a temporal holdout, between any two on-chain reputation algorithms. This thesis provides both the allowance edge and the comparison. For the single-layer methods, PageRank is taken from the literature ^[15] and applied without modification. The coupled operator modifies the random walk itself, but I show, using a pre-registered test, that the operator fails to improve upon its endpoints. My claim to new knowledge is that an on-chain reputation score's measure depends on the edges it uses, that this claim can be verified on a single cohort with a quantified degree of uncertainty, that the ERC-20-allowance edge produces a score that is cheaper to update, that differs from the transfer-based score, that reflects the intent of the edge type, that is decoupled from the transfer-based score when looking at subsequent endorsements, but that scores more poorly than the transfer-based score on measures of flow and stability, and that two additional actions that I would predict from reading the literature fail to yield better results on my dataset: propagation does not do a better job of predicting the future growth of the graph than the raw degrees that it is smoothing, and a weighted combination of the two edge types does not rank better than the best single layer. The construct-

validity framing ^{[22],[33]} that lets me make these testable claims, along with the open pipeline with published summaries of results that allows others to reproduce my analyses, form part of this work’s contribution, because verifying that decentralized financial systems are not mis-calculating credit scores is a matter of public interest ^[2].

The thesis is intentionally narrow in scope. It does not attempt to demonstrate that on-chain reputation can replace collateral ^{[1],[9]}, that EndorseRank cannot be gamed by Sybils (Section 5.6), or that its results apply outside of Arbitrum One and the time window I consider.

6.3 Future work

Each suggestion below is linked to a limitation identified in Chapter 5.

- Complete the second-tier extraction. The expanded set of addresses I used to assess node count was only used to assess the number of nodes. Extracting the approval and transfer logs for the supplemental addresses would allow me to calculate the running time of the algorithm as the number of edges increases and would transform the bounded claim for RQ3 into a general scaling claim.
- Validate the results in other contexts. Repeating the analysis on other rollups and Layer-1 networks would help establish whether the allowance-transfer tradeoff and the holdout result extend beyond Arbitrum One; the heterogeneous nature of the DeFi space ^{[1],[8]} means this is the primary threat to external validity.
- Extend the time window and use a credit-linked holdout. Extending the observation window would enable me to check whether the regression coefficients remain stable across different market conditions. Using a credit-linked holdout, where possible in the context of the lending protocols that cover the cohort of spenders, would allow me to see whether the endorsement edge captures any credit-risk information; the current holdout does not.
- Adversarial evaluation. The Sybil cost model is analytical. Empirical red-teaming with wash approvals, coordinated spender farms and strategic revocation would quantify realized attack

cost-effectiveness under live fee markets and detection heuristics ^[11], and would test whether the Sybil-adjusted stability proxies remain informative under active adversaries.

- Coupling beyond a fixed transition-level mixture. The coupled operator was tested with a fixed λ and uniform teleportation, and it did not beat its end points. Node-dependent or learned weights, mixtures of the two stationary vectors rather than of the transition rows, and per-layer damping remain untested; any such design should be pre-registered as here, report construct-specific checks and time-split evidence with degree baselines, and preserve the auditability that distinguishes graph propagation from opaque learned scores ^{[2],[14]}.
- Beating the degree. The raw t_1 degree out-predicts both PageRanks on the growth of their own construct. Whether any propagated score adds out-of-window value over the local count, on a label that is not itself a count of arrivals, is the open question that the holdout of this thesis leaves for the field.
- Richer authorization semantics. Adding protocol-specific delegation logs to the edge definition may improve coverage where ERC-20 approvals are sparse; the same reporting discipline applies.
- Deployment studies. Integration with risk oracles and governance tooling would test whether EndorseRank's runtime profile sustains production refresh cadences, and controlled fee-tier or monitoring-cadence experiments would replace the adoption hypotheses of Section 5.4.4 with evidence. Collateral reduction remains a hypothesis, not a claim of this research.

6.4 Closing reflection

On-chain reputation does not remove the need for collateral in adversarial, pseudonymous markets ^[9]. It can make behavioral structure readable, and this research shows that the ranking depends on which edges are used. Transfers record economic flow; allowances record delegated authority. PageRank can rank either graph, but the same ranking cannot serve both constructs, and one walk over both graphs serves neither better than the right single graph. EndorseRank uses the authoriza-

tion graph and reports its costs and benefits, including a time-split check of later approvals in which it separates from AWP and falls short of the raw degree, two null within-method contrasts, and a pre-registered coupling test that came out negative.

A protocol or researcher who needs a fast, auditable score of who is trusted to spend has, on Arbitrum One data of the kind studied here, a usable instrument in EndorseRank. One who needs to find transfer hubs should still use AWP. Predicting trading profit from social graphs alone is outside what either method can answer.